A Lightweight SAR Ship Detector Using End-to-End Image Preprocessing Network and Channel Feature Guided Spatial Pyramid Pooling

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***Abstract*— Recently, in the field of synthetic aperture radar (SAR) ship detection, deep-learning-based methods have made significant strides in terms of detection accuracy and speed. However, small-scale targets and complex backgrounds remain a formidable obstacle to SAR ship detection. To overcome the aforementioned challenges, this letter proposes LiteSAR-Net, a lightweight SAR ship detector, to enhance ship detection capabilities in SAR imagery. In detail, an end-to-end image preprocessing network (E2IPNet) is proposed to strengthen context information and expand the network's effective receptive field. In addition, to prevent the dilution of semantic information, the channel feature guided spatial pyramid pooling (CFGSPP) is proposed, which can adjust the parameters adaptively based on interchannel information. The proposed LiteSAR-Net achieved an average precision (AP) of 98.61% on the SAR ship detection dataset (SSDD) and 93.33% on the high-resolution SAR images dataset (HRSID), with a parameter of only 5.247 M, outperformed many state-of-the-art (SOTA) detectors. Our code can be found at https://github.com/ZYMCCX/LiteSAR-Net.**

***Index Terms*— Deep learning, lightweight, object detection, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is now widely employed in ship detection for civil and military fields [1], due to its advantages of obtaining high-resolution images and sustaining all-day, all-weather operation. To detect ships from SAR images more efficiently, deep-learning-based detectors are extensively employed. The current deep-learning-based detectors are primarily two categories: anchor-based detectors and anchor-free detectors.

The primary distinction between anchor-based and anchor-free detectors is their dependence on previously extracted anchors. As an early example of the anchor-based detector, Faster R-CNN [2] separates object detection into two stages: candidate region proposal and object regression and classification. Despite its relatively high accuracy, its computational complexity is significantly higher than that of subsequent methods. The following anchor-based methods, such as RetinaNet [3], YOLOv5, YOLOv8, and YOLOv7 [4], have only a single stage, resulting in a significant increase in computational efficiency. To overcome the aforementioned difficulties, anchor-free detectors are developed. The most recent classical anchor-free detectors, such as FCOS [5], YOLOX [6], RTMDet [7], and YOLOv8, exhibit high detection accuracy. However, when applied to SAR ship images, due to the complex background and dense ships, it is difficult to achieve high detection accuracy, with false alarms and omissions frequently occurring.

Many researchers have designed detectors for SAR ship detection by improving the classical detectors. Yang et al. [10] developed a CoAM to acquire more representative semantic features to distinguish ships from a complex background. In addition, the RFIM module is designed to collect multiscale context information. Zhang et al. [11] proposed the BL-Net to address SAR ship detection's four imbalances: image sample scene, positive negative sample, ship scale feature, and classification regression task. For instance, image sample scene imbalance may lead to a learning bias in the process of training. To eliminate the bias, BSLM is put forward to balance the learning representation. Bai et al. [12] designed the FBUA-Net. In detail, the proposed GC-FBP module is used to balance the semantic information output from various feature pyramid layers and learn the global information. Zhang and Zhang [13] proposed FL-CSE-ROIE to avoid the loss of context information and to better utilize the multiscale features extracted by FPN. Moreover, the channel attention module SENet is applied to highlight valuable features, which helps to balance contexts' contributions in different scopes. To optimize small ships detection without increasing parameter size, Chen et al. [14] constructed a CS^n-(FPN+PAN) structure referred to as CS^nNet. In particular, CS^n is introduced to build dependency relationships among pixels by extending spatial and channel features.

Although the aforementioned detectors demonstrate high accuracy in SAR ship detection, these methods are computationally inefficient because they rely on a large number of

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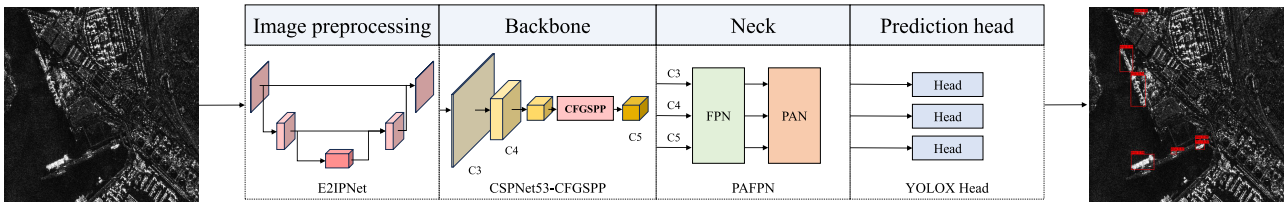


Fig. 1. Overall structure of the proposed LiteSAR-Net. The proposed E2IPNet will preprocess the input SAR images to improve the context information. Additionally, the designed CFGSPP is added to the backbone for better fusion of multiscale features and to avoid semantic information loss.

parameters to extract features. Moreover, the complexity of the background makes it difficult to distinguish ships from clutter, land buildings, and coastlines, making SAR ship detection challenging. Additionally, there are many small-scale ships in the SAR images, which makes it difficult for the backbone to extract efficient features.

Regarding SAR ship characteristics, this letter proposes LiteSAR-Net, a lightweight SAR ship detector based on YOLOX-tiny [6]. The letter's primary contributions are as follows.

1) An end-to-end image preprocessing network (E2IPNet) is proposed to enhance the context features of targets for better detection of SAR ships in complex scenes.
2) To prevent the loss of semantic information, a channel feature guided spatial pyramid pooling (CFGSPP) that can dynamically adjust its parameters by leveraging interchannel information is proposed.
3) Experiments conducted on the SAR ship detection dataset (SSDD) [8] and the high-resolution SAR images dataset (HRSID) [9] demonstrate that the proposed LiteSAR-Net achieves state-of-the-art (SOTA) performance.

II. Proposed Method

A. Overall Architecture

Considering the characteristics of ships in SAR images, a lightweight SAR ship detector, LiteSAR-Net, is proposed in this letter, and its overall architecture is depicted in Fig. 1. The proposed LiteSAR-Net consists of four parts: image preprocessing, backbone, neck, and prediction head. First, in the image preprocessing component, the input image will be preprocessed by the proposed E2IPNet to enhance the context information of the image for better detection of ships under complex backgrounds in SAR images. Subsequently, the feature maps processed by the E2IPNet will be fed into the backbone consisting of cross-stage partial network (CSPNet) and CFGSPP, which will be able to extract a set of feature maps for detecting ships at different scales. After that, for better multiscale ship detection in SAR images, neck, YOLOX-tiny's path aggregation feature pyramid network (PAFPN), will process the extracted feature map $[C_3, C_4, C_5]$. Subsequently, $[C_3, C_4, C_5]$ will pass through the prediction head to determine the ship's position in the SAR image. In YOLOX's detection head, the input feature will be refined by convolution to get three tensors Reg $\in \mathbb{R}^{4\times W\times H}$, Obj $\in \mathbb{R}^{1\times W\times H}$, and Cls $\in \mathbb{R}^{\text{Classes}\times W\times H}$. Finally, the three tensors will be stacked to get Out $\in \mathbb{R}^{(4+1+\text{Classes})\times W\times H}$, which is used to obtain the position, confidence, and class of the target.

B. End-to-End Image Preprocessing Network

Due to noise such as land structures, clutter, and coastlines in SAR images, current SAR ship detectors have difficulty locating ships in complex backgrounds. To improve ship detection with complex scenes, this letter proposes a U-shaped image preprocessing network, E2IPNet, which can enhance the ship's outline features and context information. The workflow of the designed E2IPNet is shown in Fig. 2.

At first, to extract ship features, the input $I \in \mathbb{R}^{3\times W\times H}$ will be reshaped into a $F \in \mathbb{R}^{48\times W/4\times H/4}$ by two proposed asymmetric large kernel balance blocks (ALKB-Blocks), which are capable of efficiently fusing context and local information. Subsequently, this feature map will be refined by convolutional block attention module (CBAM) [15]. Following that, to fuse the feature map, we use the ALKB-Block in conjunction with upsampling to effectively upscale the refined feature maps. Besides, to further enhance the context and position information of ships, we add the upscaled feature maps with the previous results directly. Finally, after two such operations, a processed $I' \in \mathbb{R}^{3\times W\times H}$ is obtained to be utilized as an input for the following network. The details about ALBK-Block and CBAM are shown below.

1) Asymmetric Large Kernel Balance Block: Enhanced context information can improve the detection of small ships and ships with complex backgrounds. To better merge context information with local information and to expand the effective sensing field [16], a new convolution block ALKB-Block is proposed, and its structure is presented in Fig. 2.

Large kernel convolution can effectively expand the ERF and extract more context information. However, large kernel convolution is computationally intensive and results in the loss of local information. Therefore, to balance the global and local information, we utilize a set of asymmetric convolution blocks with kernel sizes of 13 × 3 and 3 × 13 to process the feature maps. This allows for less computing complexity and a more optimal distribution of local and global knowledge. In addition, we also add a 3 × 3 convolution block to further balance the context information. Finally, the output of the ALKB-Block is the sum of the three convolutional outputs. The workflow of the designed ALKB-Block can be calculated as

$$F_{\text{ALKB}}(X) = f_{\text{ACBS}}^{3\times 13}(X) + f_{\text{ACBS}}^{13\times 3}(X) + f_{\text{CBS}}^{3\times 3}(X) \quad (1)$$

where $f_{\text{ACBS}}^{k_1\times k_2}$ denotes a convolution block, which contains an asymmetric convolution with kernel size of $k_1 \times k_2$, batch normalization, and a SiLU activation function, and $f_{\text{CBS}}^{k\times k}$ replaces the asymmetric convolution with a $k \times k$ convolution.

2) Convolutional Block Attention Module: To further highlight the ship's features in SAR images, we introduce the

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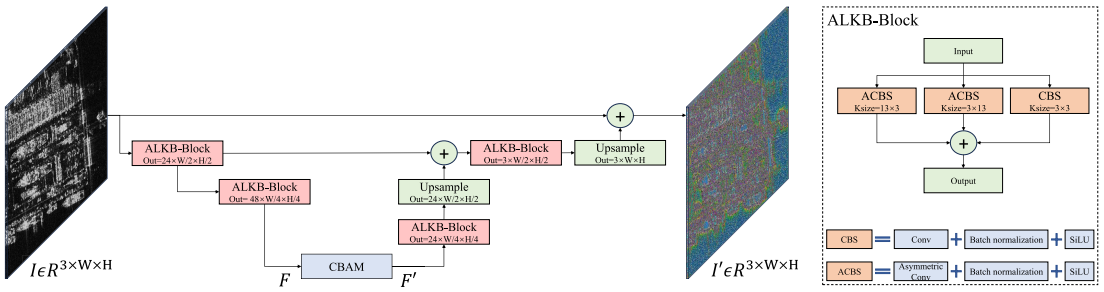


Fig. 2. Overall structure of the proposed E2IPNet and ALKB-Block.

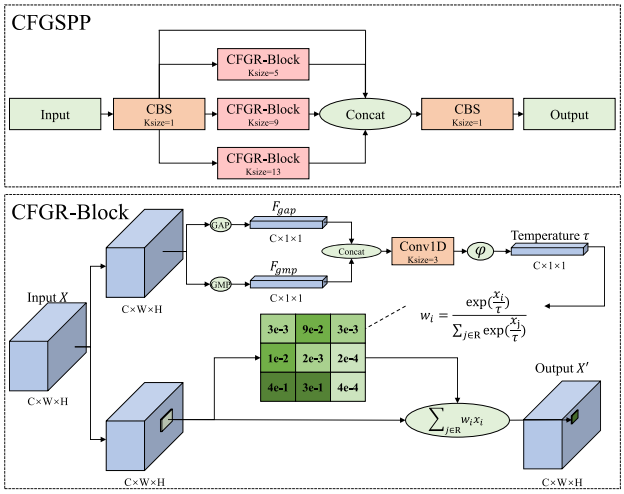


Fig. 3. Structure of the proposed CFGSPP and CFGR-Block.

CBAM, a mixed attention module, to refine the extracted feature maps. The CBAM workflow can be described as follows:

$$F_{\text{att}}(X) = f_s(f_c(X)) \tag{2}$$

$$f_c(X) = (\sigma(f(\text{GAP}(X)) + f(\text{GMP}(X)))) \times X \tag{3}$$

$$f_s(X) = \left(\sigma\left(f_{2D}^{7\times7}(\text{Concat}(\text{AP}(X), \text{MP}(X)))\right)\right) \times X \tag{4}$$

where F_{att} represents the CBAM attention module. f_c and f_s denote the channel attention submodule and spatial attention submodule, respectively. GAP and GMP represents global average pooling and global maximum pooling. AP and MP represents average pooling and maximum pooling. f stands for multilayer perceptron, while σ represents the sigmoid activation function. concat stands for the concatenate operation.

C. Channel Feature Guided SPP

YOLOX-tiny implements spatial pyramid pooling (SPP) for adjusting to images with different resolutions and fuse features at diverse scales. However, SPP extracts features at different scales via three maximum poolings with kernels $[5, 9, 13]$, which may result in the loss of semantic information [17]. Furthermore, the maximum pooling is insensitive to the features of small ships in SAR images, which may dilute the information about small ships.

To better integrate different scale features, the CFGSPP is proposed in this letter. The overall structure of the proposed CFGSPP is shown in Fig. 3. We propose a new block called channel feature guided reparametrize block (CFGR-Block), which can dynamically adjust parameters based on interchannel information. As is depicted in Fig. 3, we combine channel feature-guided temperature softmax with pooling operations to refine the feature map, which can reduce the loss of semantic information. Different from the classic temperature softmax, in which temperature is a manually designed hyperparameter, the proposed CFGR-Block can adaptively regulate the temperature through interchannel information. The process of the proposed CFGR-Block can be summarized as follows:

$$w_i = \frac{\exp\left(\frac{x_i}{\tau}\right)}{\sum_{j\in R}\exp\left(\frac{x_j}{\tau}\right)} \tag{5}$$

$$\tilde{x} = \sum_{j\in R} w_i x_i \tag{6}$$

where exp indicates the natural exponent, x_i denotes the input feature map, and τ is channel feature guided temperature. Besides, w_i is the weight of x_i, R denotes the kernel neighborhood, and $\tilde{x}$ is the refined output.

To derive the channel feature guided temperature, the input $X \in \mathbb{R}^{C\times W\times H}$ will be processed by global average pooling and global maximum pooling to generate feature maps $F_{\text{gap}} \in \mathbb{R}^{C\times1\times1}$ and $F_{\text{gmp}} \in \mathbb{R}^{C\times1\times1}$, respectively. Subsequently, F_{gap}, which contains more global information, will be concatenated with F_{gmp}, which contains more local information, to get $F' \in \mathbb{R}^{2C\times1\times1}$. The concatenated F' will be downscaled via a 1-D convolution with a kernel size of 3 and the interchannel information will be extracted to produce F_g. Finally, F_g will be passed through the proposed mapping relationship φ, thereby obtaining the temperature τ guided by channel features. This process can be formulated as follows:

$$F' = \text{Concat}(\text{GAP}(X), \text{GMP}(X)) \tag{7}$$

$$\tau = \varphi\left(f_{1D}^{3}(F')\right) \tag{8}$$

where f_{1D}^{3} denotes a 1-D convolution with kernel size 3. $\varphi(X)$ is the proposed mapping relationship.

The proposed mapping relationship should satisfy the following conditions: 1) there should be an upper limit on the temperature τ to prevent excessive smoothing and blurring of data due to a temperature that is too high; and 2) the mapping relationship should be sufficiently sensitive to input changes. The proposed mapping relationship is

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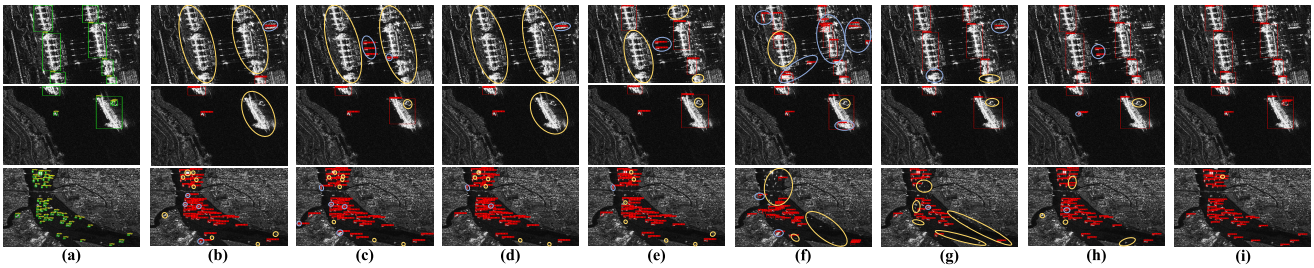


Fig. 4. Visual comparison experiment. Yellow circles indicate missed detections while blue circles are false alarms. Furthermore, (a) represents the ground truth, while (b)–(i) represent the results of YOLOv5-s, YOLOX-s, RTMDet-s, YOLOv8-s, faster R-CNN, FCOS, YOLOv7, and the proposed LiteSAR-Net.

calculated as

$$\varphi(X) = k \times \sigma(m \times (X + b)) \tag{9}$$

where σ is the sigmoid function and X represents input feature map. Among other parameters, $k = 10$, $m = 50$, and $b = -0.091$.

III. Experiments

A. Datasets and Implementation Details

1) Datasets Description: Two public challenging datasets, the SSDD and the HRSID, are introduced to validate the proposed LiteSAR-Net. SSDD is one of the most widely utilized SSDDs. This dataset contains 1160 images with varying resolutions, of which 928 images are used for training and the remaining 232 images are used for testing. HRSID is a high-resolution dataset. The HRSID dataset contains 5604 images with a resolution of 800 × 800, of which 3642 are used for training and 1962 for testing.

2) Training Details: In this letter, stochastic gradient descent (SGD) is used as the optimization algorithm, with the initial learning rate set to 0.01, the momentum set to 0.937, and the weight decay set to 0.0005. We set the batch size to 16 and trained 300 epochs. The input image sizes are 416 × 416 in SSDD and 800 × 800 in HRSID. All experiments in this letter are conducted on an Nvidia RTX 3090 GPU.

3) Evaluation Metrics: To evaluate the performance of the proposed LiteSAR-Net, average precision (AP), F1, frames per second (fps), and parameters are introduced. Whereas AP and F1 are used to evaluate detection accuracy, fps, and parameters are utilized to evaluate computational complexity and efficiency.

B. Ablation Experiments

1) Effectiveness of Each Proposed Module: In this section, experiments are conducted on SSDD and HRSID to verify the efficacy of the proposed E2IPNet and CFGSPP. As is shown in Table I, the AP of the baseline model on the SSDD and HRSID datasets is 95.08% and 90.65%, respectively, while the F1 is 0.95 and 0.89, with a parameter of 5.033 M. Compared to the baseline model, the LiteSAR-Net proposed in this letter achieves an AP for 98.61% (+3.53%) and 93.33% (+2.68%) on SSDD and HRSID, respectively, with a parameter of 5.247 M (+4.2%). With the introduction of the designed E2IPNet, the model surpasses by 2.56% and 2.32%, respectively, in terms of AP metrics on the SSDD and HRSID datasets. On the F1 metric, the addition of E2IPNet improves the baseline model by 0.01 on both datasets.

TABLE I
Ablation Experiment on the SSDD and HRSID Datasets

E2IPNet	CFGSPP	SSDD		HRSID		Parameters
		AP	F1	AP	F1	
✗	✗	95.08	0.95	90.65	0.89	5.033
✓	✗	97.64	0.96	92.97	0.90	5.246
✗	✓	98.22	0.95	93.11	0.91	5.034
✓	✓	98.61	0.96	93.33	0.91	5.247

TABLE II
Effectiveness of the Proposed CFGR-Block

Method	AP	F1	Parameters
Maximum Pooling (baseline)	90.65	0.89	5.033
Average pooling	89.88	0.87	5.033
Convolution	90.66	0.88	15.171
CFGR-Block (ours)	93.11	0.91	5.034

2) Effectiveness of the Proposed CFGR-Block: To validate the proposed CFGR-Block further, the maximum pooling of the baseline model is replaced sequentially by average pooling, convolution, and the proposed CFGR-Block with the same kernel size. As shown in Table II, the proposed CFGR-Block outperformed the baseline's maximum pooling, average pooling, and convolution by 2.46%, 3.23%, and 2.45%, respectively. Moreover, the parameters of CFGR-Block are very close to the maximum pooling and average pooling, which is only 33.1% of the convolution.

C. Comparison Experiments

In this section, we conduct comparative experiments to further validate the effectiveness of the proposed LiteSAR-Net. LiteSAR-Net is compared with both lightweight and classical SOTA detectors. For lightweight detectors, we include YOLOv5-s, YOLOX-s [6], RTMDet-s [7], and YOLOv8-s in our comparison. As for classical SOTA detectors, our comparison encompasses Faster R-CNN [2], YOLOv7 [4], and FCOS [5]. Furthermore, we compare LiteSAR-Net with the latest SAR ship detectors, such as the method proposed by Yang et al. [10], BL-Net [11], FBUA-Net [12], FL-CSE-ROIE [13], and CS^{n}Net [14]. The results of these comparisons are presented in Table III.

Among these classic SOTA detectors with the highest detection accuracy is YOLOv7, which has an AP of 97.22% on the SSDD and 92.85% on the HRSID datasets, respectively. In contrast, LiteSAR-Net is 1.39% and 0.48% higher than YOLOv7 in SSDD and HRSID for AP, respectively, whereas the number of parameters is only 14.1% of YOLOv7. Among

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TABLE III
COMPARISON WITH THE SOTA OBJECT DETECTORS AND SAR SHIP DETECTORS ON THE SSDD AND HRSID DATASETS

Method	SSDD		HRSID		Parameters
	AP	F1	AP	F1	
YOLOv5-s	95.89	0.91	88.32	0.83	7.064
YOLOX-s	96.06	0.94	90.39	0.87	8.938
RTMDet-s	92.80	0.91	89.26	0.86	8.892
YOLOv8-s	95.89	0.91	90.56	0.86	11.136
Faster R-CNN	89.74	0.88	85.95	0.81	136.689
FCOS	96.46	0.92	88.43	0.83	32.111
YOLOv7	97.22	0.94	92.85	0.88	37.192
Yang et al.[10]*	96.4	0.94	91.3	0.86	73.5
BL-Net[11]*	95.25	0.93	88.67	0.90	47.81
FBUA-Net[12]*	96.2	-	90.3	-	36.54
FL-CSE-ROIE[13]*	95.9	-	90.2	-	107.55
CS^{n}Net[14]*	97.1	-	91.2	-	12.2
LiteSAR-Net (ours)	**98.61**	**0.96**	**93.33**	**0.91**	**5.247**

* Denotes data obtained from their papers.

these SAR ship detectors, CS^{n}Net has the minimum computational complexity, with a parameter that is 6.953 M more than the proposed LiteSAR-Net. Furthermore, the proposed LiteSAR-Net outperforms CS^{n}Net by 2.13% in terms of AP metrics.

Besides, visual comparison experiments are also conducted, and their results are shown in Fig. 4. As illustrated in Fig. 4, in the scenarios depicted in the first column of the figure, almost all detectors identified the reflection at the image center as ships, resulting in numerous false positives. Only LiteSAR-Net and FCOS avoided false detections in this regard. However, FCOS missed detecting some of the ships on shore, while LiteSAR-Net successfully detected all the ships in the image. Additionally, in the scenario depicted in the third column of the figure, featuring numerous dense ships, the detection accuracy of LiteSAR-Net, despite missing a small ship, notably surpasses that of other detectors like YOLOv7.

IV. CONCLUSION

This letter proposes LiteSAR-Net, an innovative lightweight SAR ship detector, with the intent of enhancing ship detection capabilities in SAR imagery. On the one hand, E2IPNet is designed to expand the network's effective receptive field and to improve the context information of ships in SAR images. CFGSPP, on the other hand, is proposed to reduce the loss of semantic information and to fuse multiscale ship features. Experimental results demonstrate that the proposed LiteSAR-Net outperforms other existing SOTA methods in terms of precision and efficiency. However, horizontal bounding boxes-based object detection introduces more clutter regions and is prone to causing false negatives in postprocessing. In our future work, we will attempt to introduce oriented bounding boxes-based object detectors to tackle this challenge at its core.

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